

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE																																													
Supplementary Examination Summer 2024																																													
Course: B. Tech. Branch : AI&DS and Allied branches Semester : V																																													
Subject Code & Name: BTAIC502 Machine Learning																																													
Max Marks: 60 Date:03/07/2024 Duration: 3 Hr.																																													
Instructions to the Students:																																													
1. All the questions are compulsory.																																													
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.																																													
3. Use of non-programmable scientific calculators is allowed.																																													
4. Assume suitable data wherever necessary and mention it clearly.																																													
5. Use of Z table, T table and F-distribution table is allowed.																																													
										(Level/CO)	Marks																																		
Q. 1	Solve Any Two of the following.											12																																	
A)	Define Machine Learning. Explain the any two types of machine Learning with suitable diagram.										(BT2/CO1)	6																																	
B)	Explain the categories of Machine Learning based on required output, including classification, Regression and Clustering.										(BT1/CO1)	6																																	
C)	Explain the steps of data preparation in machine learning, such as reading data file, pre-processing, and splitting data.										(BT4/CO1)	6																																	
Q.2	Solve Any Two of the following.											12																																	
A)	List and explain Performance Metrics for Regression in detail.										(BT2/CO2)	6																																	
B)	<table><tr><td>N=200</td><td colspan="5">Prediction "NO"</td><td colspan="5">Prediction "YES"</td></tr><tr><td>Actual =NO</td><td colspan="5">60</td><td colspan="5">10</td></tr><tr><td>Prediction =YES</td><td colspan="5">5</td><td colspan="5">125</td></tr></table> Using above confusion matrix, calculate the following 1.Accuracy 2.Error Rate 3.Precision 4.Recall 5. Specificity 6.F1 Score										N=200	Prediction "NO"					Prediction "YES"					Actual =NO	60					10					Prediction =YES	5					125					(BT2/CO2)	6
N=200	Prediction "NO"					Prediction "YES"																																							
Actual =NO	60					10																																							
Prediction =YES	5					125																																							
C)	Compare Classification Report and AUC (Area Under ROC Curve) with one example. Assume suitable data for example.										(BT4/CO2)	6																																	
Q. 3	Solve Any Two of the following.											12																																	
A)	Define Gradient Descent. Explain different variations of Gradient Descent.										(BT2/CO3)	6																																	
B)	What is Linear Regression? Explain the steps in linear Regression.										(BT2/CO3)	6																																	
C)	For data given below: <table><tr><td>x</td><td>0</td><td>1</td><td>2</td><td>3</td><td>4</td><td>5</td><td>6</td><td>7</td><td>8</td><td>9</td></tr><tr><td>y</td><td>2</td><td>2</td><td>3</td><td>5</td><td>6</td><td>8</td><td>9</td><td>10</td><td>11</td><td>13</td></tr></table> a) Find the least square regression line $y=B_1x + B_0$ b) Estimate the value of y when x=10										x	0	1	2	3	4	5	6	7	8	9	y	2	2	3	5	6	8	9	10	11	13	(BT2/CO3)	6											
x	0	1	2	3	4	5	6	7	8	9																																			
y	2	2	3	5	6	8	9	10	11	13																																			

Q.4	Solve Any Two of the following.		12										
A)	Explain the Random Forest Algorithm with examples.	(BT4/CO4)	6										
B)	What is the purpose of the coefficient of determination (R-Squared) in linear regression and how is it calculated?	(BT2/CO4)	6										
C)	A bank wants to predict the likelihood of a loan being approved based on the credit score of the applicant. The data is as follows:<table><tr><th>Credit Score</th><th>Loan Approved</th></tr><tr><td>700</td><td>1</td></tr><tr><td>650</td><td>0</td></tr><tr><td>750</td><td>1</td></tr><tr><td>600</td><td>0</td></tr></table>Using logistic regression, find the optimal coefficients and predict the likelihood of a loan being approved for an applicant with a credit score of 720.	Credit Score	Loan Approved	700	1	650	0	750	1	600	0	(BT2/CO4)	6
Credit Score	Loan Approved												
700	1												
650	0												
750	1												
600	0												
Q. 5	Solve Any Two of the following.		12										
A)	Explain the working of KNN algorithm with example?	(BT1/CO5)	6										
B)	Explain Naïve Bayes classifier with an example?	(BT2/CO5)	6										
C)	Explain the following: 1. Support Vector Classifier 2. Support Vector Regression	(BT2/CO5)	6										
	*** End ***												